

Algorithms 2

Minimum Set-Cover

Daniel Rosenberg & Michael Trushkin

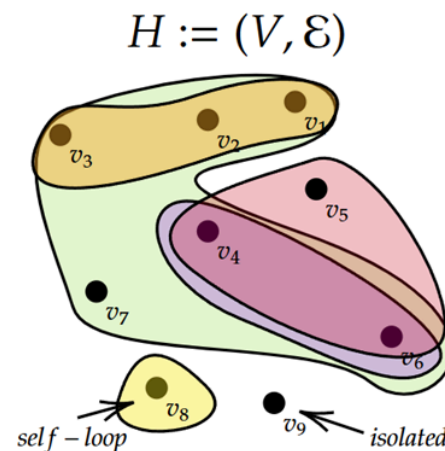
Ariel University

1. The Minimum Set-Cover Problem

Definition 1 Hypergraph

By a **hypergraph** we mean an ordered pair $H := (V, \mathcal{E})$ with $\mathcal{E} \subseteq \mathcal{P}(V)$.

- If all members of $\mathcal{E}$ are **pairs**, then H is called a **graph**
- Members of $\mathcal{E}$ are called **hyper-edges**



$$V = \{v_1, \dots, v_9\}$$

$$\mathcal{E} = \{e_1, e_2, e_3, e_4, e_5\}$$

$$e_1 = \{v_1, v_2, v_3\}$$

$$e_2 = \{v_1, v_2, v_3, v_4, v_5\}$$

$$e_3 = \{v_4, v_6\}$$

$$e_4 = \{v_4, v_5, v_6\}$$

$$e_5 = \{v_8\}$$

1.1 Hypergraphs

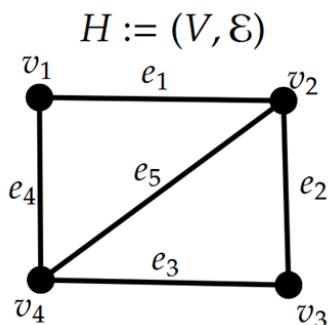
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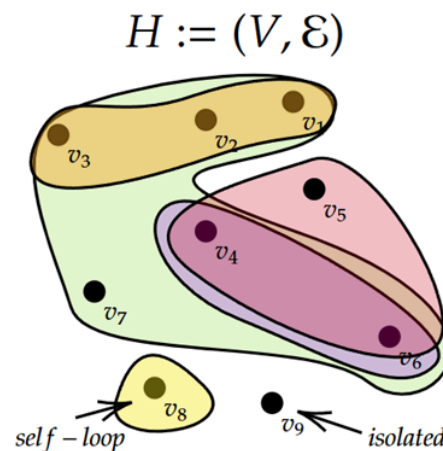
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- Members of $\mathcal{E}$ are called **hyper-edges**

FYI: If $|e| = k$ for all $e \in \mathcal{E}$, then H is called **k -uniform**.

- A graph is a **2-uniform** hypergraph.



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 $\mathcal{E} = \{e_1, e_2, e_3, e_4, e_5\}$
 $e_1 = \{v_1, v_2\}$
 ...



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Problem 3 Minimum Set-Cover (SC)

Instance: A hypergraph $H := (V, \mathcal{E})$.

Goal: A subset $C \subseteq \mathcal{E}$ such that $\bigcup_{e \in C} e = V$, minimising $|C|$.

Put another way: We seek a minimum **hyper-edge-cover** of H .

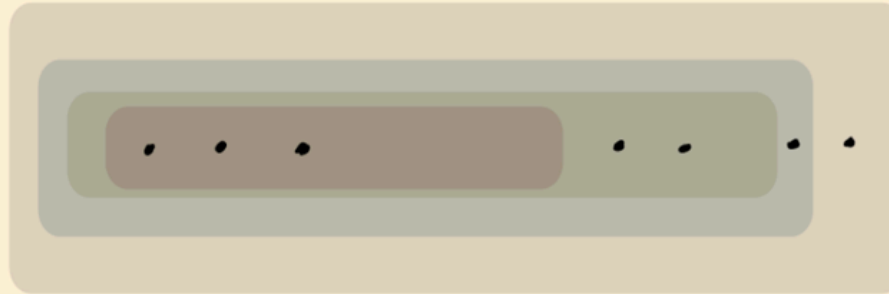
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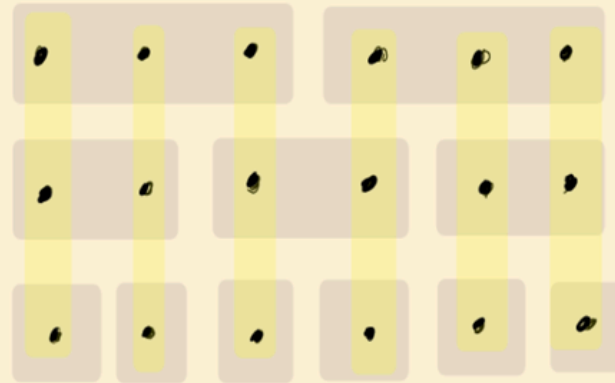
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- Recall: for graphs, the minimum **edge-cover** problem is in P . (Find maximum matching and extend greedily)
- The minimum SC problem is **NP-complete** (we return to this below)



optimal sol. has size 1.



optimal sol. in yellow

$\text{VC} := \{(G, k) : \tau(G) \leq k\}$
min vertex cover problem

$\text{SC} := \{(H, k) : H \text{ has a hyper-edge cover of size } \leq k\}$

Claim: $\text{VC} \leq_p \text{SC}$.

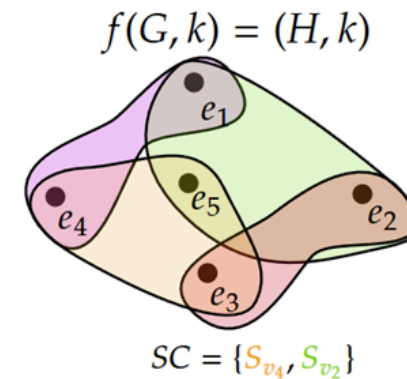
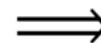
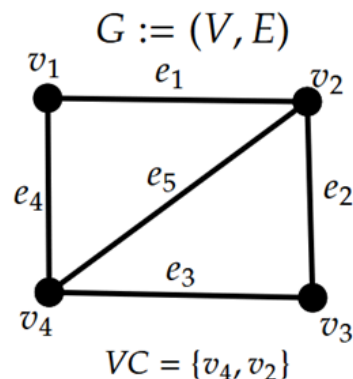
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Claim: $\text{VC} \leq_p \text{SC}$.

For each $u \in V(G)$, define: $S_u := \{e \in E(G) : u \in e\}$.

Given (G, n) , define a hypergraph H by:

$$V(H) := E(G), \quad \mathcal{E}(H) := \{S_u : u \in V(G)\}$$



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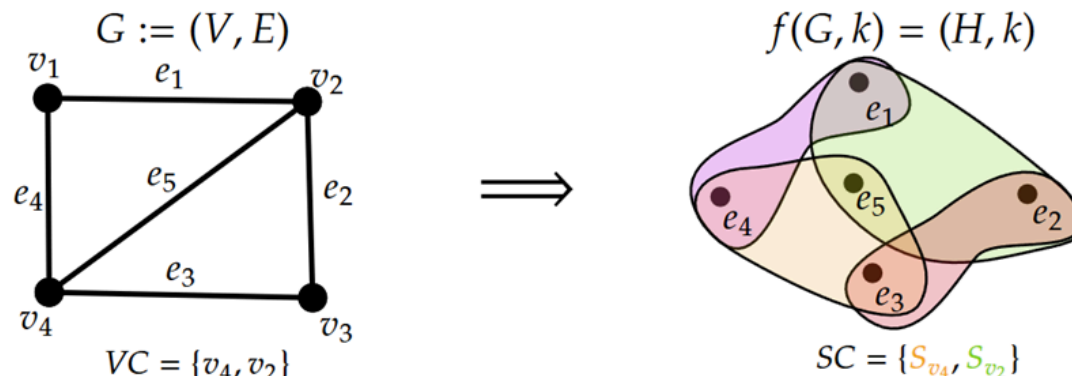
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Question 3 H.W. Complete the argument.



2. Greedy Algorithm for Minimum Set-Cover

Notation: For a set-system A , write $\bigcup A := \bigcup_{e \in A} e$.

Algorithm 1: Greedy-SC

```
1:  $A \leftarrow \emptyset$ 
2: while  $\bigcup A \neq V$  do
3:    $e \leftarrow \arg \max\{|e \setminus \bigcup A| : e \in \mathcal{E}\}$ 
4:    $A \leftarrow A \cup \{e\}$ 
5: end
6: return  $A$ 
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(hyper-edge covering the most new vxs)

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Algorithm 2: Greedy-SC

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Question 5 What is the approximation ratio of the greedy algorithm?

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Trivially $m \leq |V(H)| = n$.

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Theorem 2 The greedy algorithm is an $O(\ln m)$ -approximation for minimum set-cover.

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Observation 8 Size of the cover output = # iterations greedy performs.

Define: A_i = greedy solution at the **beginning** of the i -th iteration.

- Greedy transitions $A_i \rightarrow A_{i+1}$ during iteration i .
- # new vxs covered = $|\bigcup A_{i+1}| - |\bigcup A_i|$.

(all from $V(H) \setminus \bigcup A_i$)

Let OPT = size of an optimal cover.

Claim : An optimal solution $P := \{e_1, \dots, e_{\text{OPT}}\}$ contains e_j s.t.:

$$|e_j \cap (V(H) \setminus \bigcup A_i)| \geq \frac{|V(H) \setminus \bigcup A_i|}{\text{OPT}}$$

Proof (By contradiction):

- Assume no such e_j exists

$$\Rightarrow \forall e \in \mathcal{E} : |e \cap (V(H) \setminus \bigcup A_i)| < \frac{|V(H) \setminus \bigcup A_i|}{\text{OPT}}$$

Then,

$$\underbrace{|V(H)| = |\bigcup P|}_{P \text{ cover all of } H} < \text{OPT} \cdot \frac{|V(H) \setminus \bigcup A_i|}{\text{OPT}} = \underbrace{|V(H) \setminus \bigcup A_i|}_{(V(H) \setminus \bigcup A_i) \subseteq V(H)} \leq |V(H)|.$$

Since A_i is constructed **greedily**:

Observation 9 $\forall i \in \mathbb{N} : \quad \left| \bigcup A_{i+1} \right| - \left| \bigcup A_i \right| \geq \frac{|V(H) \setminus \bigcup A_i|}{\text{OPT}}$

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Q: Once A_{i+1} is defined, how many vxs are left to cover?

$$\begin{aligned} |V(H) \setminus \bigcup A_{i+1}| &= |V(H)| - |\bigcup A_{i+1}| \\ &\leq |V(H)| - |\bigcup A_i| - \frac{|V(H) \setminus \bigcup A_i|}{\text{OPT}} \\ &= \left(1 - \frac{1}{\text{OPT}}\right) |V(H) \setminus \bigcup A_i| \end{aligned}$$

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Observation 14 $|V(H) \setminus \bigcup A_{i+1}| \leq \left(1 - \frac{1}{\text{OPT}}\right) |V(H) \setminus \bigcup A_i|$

After each iteration, the # of left-over vxs reduces by a factor of $(1 - 1/\text{OPT})$.

- Set $U_i := V(H) \setminus \bigcup A_i$
- Initially $U_0 = V(H)$:

(left-over vxs at iter i)

$$|U_{i+1}| \leq \left(1 - \frac{1}{\text{OPT}}\right) |U_i|$$

By induction: $\forall k \geq 0$:

$$|U_k| \leq \left(1 - \frac{1}{\text{OPT}}\right)^k \cdot |U_0| \leq \exp\left(-\frac{k}{\text{OPT}}\right) \cdot n$$

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The sequence $U_0, U_1, \dots$ reaches zero when $\exp\left(-\frac{i}{\text{OPT}}\right) \cdot n < 1$.

i.e.

$$i > \text{OPT} \cdot \ln n.$$

After $\text{OPT} \cdot \ln n$ iterations, all of $V(H)$ is covered.

$$\begin{aligned} \exp\left(-\frac{i}{\text{OPT}}\right) \cdot n &< 1 \\ \ln\left(\exp\left(-\frac{i}{\text{OPT}}\right) \cdot n\right) &< \ln(1) \\ -\frac{i}{\text{OPT}} + \ln n &< 0 \\ n &< \frac{i}{\text{OPT}} \end{aligned}$$

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After $\text{OPT} \cdot \ln n$ iterations, all of $V(H)$ is covered.

Recalling that size of greedy sol = # iterations:

Theorem 6 size of greedy sol $\leq (\ln n) \cdot \text{OPT}$ — an $O(\ln n)$ -approximation.

$$\begin{aligned} \exp\left(-\frac{i}{\text{OPT}}\right) \cdot n &< 1 \\ \ln\left(\exp\left(-\frac{i}{\text{OPT}}\right) \cdot n\right) &< \ln(1) \\ -\frac{i}{\text{OPT}} + \ln n &< 0 \\ n &< \frac{i}{\text{OPT}} \end{aligned}$$

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Choose i^* as the last iteration where $\text{OPT} \leq |U_{i^*}|$.

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Estimating i^* :

$$\text{OPT} \leq |U_{i^*}| \leq \left(1 - \frac{1}{\text{OPT}}\right)^{i^*} \cdot |U_0| \leq \exp(-i^*/\text{OPT}) \cdot n.$$

$$\exp(i^*/\text{OPT}) \leq \frac{n}{\text{OPT}} \implies i^* \leq \text{OPT} \cdot \ln\left(\frac{n}{\text{OPT}}\right).$$

$$\implies \text{size of greedy sol} \leq \text{OPT} \cdot \ln\left(\frac{n}{\text{OPT}}\right) + \text{OPT} = \left(1 + \ln\left(\frac{n}{\text{OPT}}\right)\right) \cdot \text{OPT}.$$

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Observation 18 $m = \max\{|e| : e \in \mathcal{E}\} \geq \frac{n}{\text{OPT}}$ since every optimal solution must contain a hyperedge of size $\geq n/\text{OPT}$.

$$\text{size of greedy sol} \leq (1 + \ln m) \cdot \text{OPT} = O(\ln m) \cdot \text{OPT}. \quad \square$$

3. Minimum Cost Set-Cover

Problem 6 Minimum Cost Set-Cover

Instance: A hypergraph H ; a cost function $c : \mathcal{E}(H) \rightarrow \mathbb{R}_{\geq 0}$.

Goal: An edge-cover $C \subseteq \mathcal{E}(H)$ of H minimising $\sum_{e \in C} c(e)$.

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The cardinality greedy **fails** for the cost/weight version. Can we still learn from it?

Consider the i -th iteration (with $V(H) \setminus \bigcup A_i \neq \emptyset$).

The cardinality greedy picks $e \in \mathcal{E}(H) \setminus A_i$ maximising:

$$\frac{|e \setminus \bigcup A_i|}{|V(H) \setminus \bigcup A_i|} = \frac{\# \text{ new vxs covered by } e}{\# \text{ left-over vxs}}$$

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For weighted: we would want an edge $e \in \mathcal{E}(H) \setminus A_i$ maximising:

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Back to cardinality: For $e \in \mathcal{E}(H) \setminus A_i$, consider $1/|e \setminus \bigcup A_i|$.

- The **smaller** this ratio, the **more** new elements e covers.

For weighted: $c(e)/|e \setminus \bigcup A_i|$ – distribute cost $c(e)$ among newly covered vxs. (This is amortisation!)

Pick $e \in \mathcal{E}(H) \setminus A_i$ minimising

$$\frac{c(e)}{|e \setminus \bigcup A_i|} = \frac{\text{cost of } e}{\# \text{ vxs covered by } e}$$

Definition 8 Effectiveness

Let $U \subseteq V(H)$ and $e \in \mathcal{E}(H)$. Define:

$$\text{eff}_U(e) := \begin{cases} \infty & \text{if } e \subseteq U \\ \frac{c(e)}{|e \setminus U|} & \text{otherwise} \end{cases}$$

to be the **effectiveness** of e with respect to U .

Algorithm 3: Greedy-Cost-SC

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Algorithm 4: Greedy-Cost-SC

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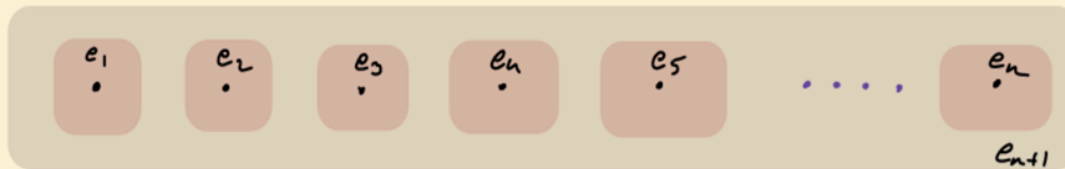
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```

Theorem 8 The greedy algorithm above forms an $O(\ln n)$ -approximation for minimum cost set-cover.

∴ An example for how the algo "works"

$H :=$



$$c(e_1) = \frac{1}{n} \quad c(e_2) = \frac{1}{n-1} \quad c(e_3) = \frac{1}{n-2} \quad \dots \quad c(e_n) = \frac{1}{n-(n-1)} = 1$$

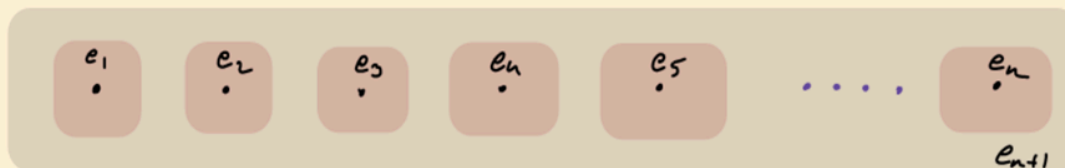
$$c(e_{n+1}) = 1 + \varepsilon, \quad \varepsilon > 0 \text{ predetermined}$$

OPTIMAL SOLUTION:

Take e_{n+1} only with cost $1 + \varepsilon$

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OPTIMAL SOLUTION:

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Greedy (step 1, $A = \emptyset$):

- $\text{eff}_{\emptyset}(e_i) = c(e_i)/1.$
- $\text{eff}_{\emptyset}(e_{n+1}) = (1 + \varepsilon)/n.$
 $\Rightarrow e_1$ chosen.

(eff = $1/n$, the smallest)

Greedy (step 2, $A = \{e_1\}$):

- $\text{eff}_{\{e_1\}}(e_i) = c(e_i)/1$
- $\text{eff}_{\{e_1\}}(e_{n+1}) = (1 + \varepsilon)/(n - 1)$
 $\Rightarrow e_2$ chosen.

(eff = $\frac{1}{n-1}$, the smallest)

...

Greedy selects $e_1, e_2, \dots, e_n$ with total cost:

$$\sum_{i=1}^n c(e_i) = \sum_{i=1}^n \frac{1}{n+1-i} = \sum_{j=1}^n \frac{1}{j} \approx \ln n \quad (\text{OPT} = 1 + \varepsilon)$$

4. Analysis of Weighted Greedy

Claim : Let P be an optimal cover with cost OPT . Then $\exists e \in P$ such that $c(e)/|e| \leq \text{OPT}/n$.

Proof. Suppose $c(e)/|e| > \text{OPT}/n$ for all $e \in P$. Then:

$$\text{OPT} = \sum_{e \in P} c(e) > \sum_{e \in P} |e| \cdot \frac{\text{OPT}}{n} = \underbrace{\frac{\text{OPT}}{n} \cdot \sum_{e \in P} |e|}_{\substack{\sum_{e \in P} |e| \geq |\bigcup_{e \in P} e| = n \\ P \text{ covers all } n \text{ vxs.}}} \geq \frac{\text{OPT}}{n} \cdot n = \text{OPT}. \quad .$$

□

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□

Amortised perspective:

Observation 20 $\exists e \in P$ covering its vxs at amortised cost $\leq \text{OPT}/n$ **per vertex**.

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This argument persists at **later stages**. Let P = optimal cover, A_i = current greedy cover. Define:

$$T := \{e \in P : e \setminus \bigcup A_i \neq \emptyset\} \quad (\text{All hyper-edges in } P \text{ that cover "new" vertices})$$

- Only members of T are relevant to the greedy: $\text{eff}_{\bigcup A_i}(e) < \infty$ iff $e \in T$
- $(V(H) \setminus \bigcup A_i) \subseteq \bigcup T$ (exercise: proof by contradiction, i.e. remaining vertices are covered by T .)

Lemma 9 (Key Claim) $\exists e \in T$ s.t.

$$\text{eff}_{\bigcup A_i}(e) = \frac{c(e)}{|e \setminus \bigcup A_i|} \leq \frac{\text{OPT}}{|V(H) \setminus \bigcup A_i|}.$$

Lemma 10 (Key Claim) $\exists e \in T$ s.t.

$$\text{eff}_{\bigcup A_i}(e) = \frac{c(e)}{|e \setminus \bigcup A_i|} \leq \frac{\text{OPT}}{|V(H) \setminus \bigcup A_i|}.$$

Proof. Suppose not. Then:

$$\begin{aligned} \text{OPT} &\geq \sum_{e \in T} c(e) > \frac{\text{OPT}}{|V(H) \setminus \bigcup A_i|} \cdot \sum_{e \in T} |e \setminus \bigcup A_i| \\ &\geq \underbrace{\frac{\text{OPT}}{|V(H) \setminus \bigcup A_i|} \cdot |V(H) \setminus \bigcup A_i|}_{\sum_{e \in T} |e \setminus \bigcup A_i| \geq |\bigcup (T \setminus \bigcup A_i)| = |V(H) \setminus \bigcup A_i|} = \text{OPT}. \quad \square \end{aligned}$$

Augment the greedy to assign a **price** to each vertex when first covered:

Algorithm 5: Greedy-Cost-SC (augmented)

```
1:  $A \leftarrow \emptyset$ 
2: while  $\bigcup A \neq V$  do
3:    $e \leftarrow \arg \min \{ \text{eff}_{\bigcup A}(e) : e \in \mathcal{E}(H) \}$ 
4:   for each  $v \in e \setminus \bigcup A$  do
5:     Set  $\alpha(v) := \text{eff}_{\bigcup A}(e)$ 
6:   end
7:    $A \leftarrow A \cup \{e\}$ 
8: end
9: return  $A$ 
```

Reformulation: Choose $e \in \mathcal{E}(H)$ whose avg cost per **newly covered** element is least.

Lemma 11 (Cost) cost of greedy sol = $\sum_{v \in V} \alpha(v)$.

Proof. Let $A = \{e_1, \dots, e_k\}$ be the greedy sol; $A_j = \{e_1, \dots, e_j\}$ the partial sol at step j .

$$\sum_{e \in A} c(e) = \sum_{j=1}^k c(e_j) = \sum_{j=1}^k \sum_{v \in e_j \setminus \bigcup_{i < j} A_i} \underbrace{\frac{c(e_j)}{|e_j \setminus \bigcup_{i < j} A_i|}}_{=\alpha(v)} = \sum_{v \in V} \alpha(v). \quad \square$$

Theorem 12 The weighted greedy algorithm is an $O(\ln n)$ -approximation.

Proof. Order $V = \{v_1, v_2, \dots, v_n\}$ by the order the greedy covers them.

When v_i is covered at iteration j , there are $\geq n - i + 1$ uncovered vxs. By **Lemma 10**:

$$\alpha(v_i) = \text{eff}_{\bigcup A_j}(e_j) \leq \frac{\text{OPT}}{|V(H) \setminus \bigcup A_j|} \leq \frac{\text{OPT}}{n - i + 1}.$$

Theorem 13 The weighted greedy algorithm is an $O(\ln n)$ -approximation.

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Therefore, using **Lemma 11**

$$\text{cost of greedy sol} = \sum_{v \in V} \alpha(v) = \sum_{i=1}^n \alpha(v_i) \leq \text{OPT} \cdot \sum_{i=1}^n \frac{1}{n - i + 1} = \text{OPT} \cdot H_n.$$

where $H_n = \sum_{i=1}^n 1/i \approx \ln n$ is the n -th **Harmonic number**. $\square$